NATURAL LANGUAGE PROCESSING · LESSON 5

Reading Feelings with VADER

How a dictionary and a few clever rules can measure the emotion in text.



WARM-UP

Rate these words

On a scale from **−4** (very negative) to **+4** (very positive), how would you rate each word?

love | good | okay | meh | boring | hate

Try it with a partner

Write your scores, then compare. Did you agree on every word? Which words were hardest to rate?

You just did real research!

This is exactly how VADER's dictionary was built. Real people rated thousands of words on this same scale.

WHY THIS MATTERS

Who wants to measure feelings in text?



Product reviews

Shops read thousands of reviews to learn what customers love and what they hate.



School feedback

Schools can quickly see how students feel about classes, events, and the canteen.



Social media

Brands and news teams track how people feel about a topic as it trends.



Customer service

Angry messages can be sorted to the top, so they get help first.

WHY THIS MATTERS

Too many opinions to read

The problem

10,000+

comments, reviews, or survey answers. Reading them all by hand would take days.

The goal

seconds

A computer reads them all and gives each one a feeling score.

This task is called **sentiment analysis**: finding out if a text is positive, negative, or neutral.

BIG PICTURE

Two ways to detect feelings

Way 1: Learn from examples

Train a model on thousands of labeled texts, like our spam detector (Lesson 2) or the transformer (Lesson 4).

Needs labeled training data.

Way 2: Dictionary + rules

Use a list of words with feeling scores, plus rules for things like “not,” “very,” and “!!!”. This is **VADER**.

No training needed. Works right away.

Today we explore Way 2, build our own mini version, and test when it works best.

LEARNING GOALS

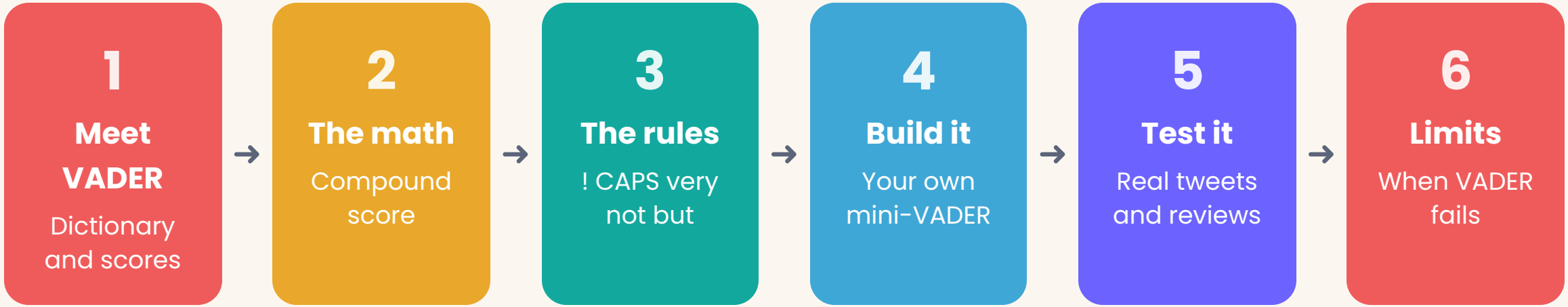
What we will learn today

1. Explain what sentiment analysis is and why it is useful.
2. Describe VADER’s two parts: a dictionary and rules.
3. Calculate a compound score by hand.
4. Predict how “!”, CAPS, “very,” “not,” and “but” change a score.
5. Build and test your own mini-VADER in Python.
6. Judge when VADER works well and when it fails.

ROADMAP

Today’s journey

Here is our plan for today’s lesson.



01

Meet VADER

A sentiment tool made for social media.

MEET VADER

What is VADER?

VADER stands for **V**alence **A**ware **D**ictionary and s**E**ntiment **R**easoner. It was created by C.J. Hutto and Eric Gilbert in 2014.

Valence means how positive or negative a word feels.

It was designed for **social media**, so it understands slang, emoticons, and emojis.

It is **free**, very **fast**, and needs **no training**.

Part 1: the dictionary (lexicon)

Word	Score	Word	Score
love	+3.2	okay	+0.9
great	+3.1	meh	−0.3
amazing	+2.8	boring	−1.3
:)	+2.0	:(	−1.9
good	+1.9	bad	−2.5
lol	+1.8	hate	−2.7

VADER's dictionary, called a **lexicon**, has **7,506** words, slang terms, and emoticons.

Each one was rated by **10 people** from -4 to +4. The score is their average.

Words that are not in the lexicon, like "table," count as neutral.

MEET VADER

Part 2: the rules

!

Exclamation points make feelings stronger.

CAPS

Writing a word in capitals adds emphasis.

very / kind of

Degree words make feelings stronger or weaker.

not

Negation flips the feeling the other way.

but

The part after “but” matters more.

😍 :) lol

Emojis, emoticons, and slang carry feelings too.

The rules are what make VADER smarter than just adding up word scores.

MEET VADER

VADER's four scores

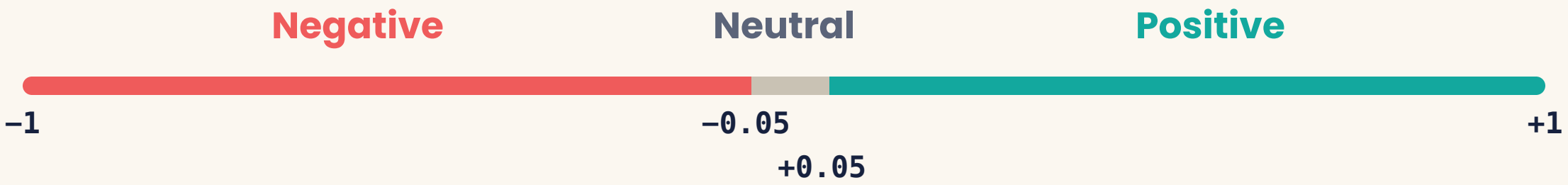
For every text, VADER gives four numbers.

Score	Meaning	Example: "The food was good."
`pos`	The share of the text that feels positive	`0.492`
`neu`	The share of the text that feels neutral	`0.508`
`neg`	The share of the text that feels negative	`0.0`
`compound`	**One overall score from -1 to +1**	**`0.4404`**

pos, neu, and neg add up to 1. Most of the time, we only need the **compound** score.

From a score to a label

VADER's creators recommend these cut-offs for the compound score.



Positive

compound ≥ 0.05

Neutral

between -0.05 and 0.05

Negative

compound ≤ -0.05

SIMULATED DEMO

Your first VADER score

SIMULATED DEMO

```
from vaderSentiment.vaderSentiment import (
    SentimentIntensityAnalyzer)

analyzer = SentimentIntensityAnalyzer()

scores = analyzer.polarity_scores("The food was good.")
print(scores)
```

OUTPUT

```
{'neg': 0.0, 'neu': 0.508, 'pos': 0.492, 'compound':
0.4404}
```

We install VADER with one command: **pip install vaderSentiment**.

SentimentIntensityAnalyzer loads the lexicon and the rules.

polarity_scores() returns the four scores for any text.

02

The Math

How VADER calculates the compound score.

THE MATH

Calculating a score step by step

Let’s calculate the score for “The food was good.” by hand.

Step 1. Find words in the lexicon. Only “good” is there, with a score of **1.9**.

Step 2. Apply the rules. There is no “!”, CAPS, “very,” “not,” or “but.”

Step 3. Add up the scores. The total is **1.9**.

Step 4. Squeeze the total into the range -1 to +1.

```
compound = total ÷ √(total² + 15)

= 1.9 ÷ √(1.9² + 15)
= 1.9 ÷ √18.61
= 1.9 ÷ 4.314
= 0.4404 → positive
```

THE MATH

Why we squeeze the score



The total can be any size, like 1.9, or 10, or -25.

The formula **total ÷ √(total² + 15)** squeezes every total into a number between -1 and +1.

The curve gets flatter near ±1. Adding more positive words still helps, but less and less.

03

The Rules

Small details that change the feeling.

RULE 1

Exclamation points add energy

Sentence	Compound score
The food was good.	`0.4404`
The food was good!	**`0.4926`**
The food was good!!!	**`0.5826`**

Each “!” adds **0.292** to the total, up to 4 exclamation points.

Check: 1.9 + 3 × 0.292 = 2.776, and 2.776 ÷ √(2.776² + 15) = **0.5826**.

RULE 2

CAPS shows strong emotion

Sentence	Compound score
The food was good!!!	`0.5826`
The food was GOOD!!!	**`0.6714`**

A word in ALL CAPS gets **0.733** more feeling, but only when the other words are not all in caps.

Why? If someone writes everything in caps, it is just their style, not extra emotion.

RULE 3

Degree words make feelings stronger or weaker

Sentence	Compound score
The food was kind of good.	`0.3832`
The food was good.	`0.4404`
The food was very good.	**`0.4927`**
The food was extremely good.	**`0.4927`**

"Very" and "extremely" are **boosters**. They add **0.293**.

"Kind of" and "slightly" are **dampeners**. They take away **0.293**. VADER knows 84 of these degree words.

RULE 4

Negation flips the feeling

Sentence	Compound score
The food was good.	**+0.4404**
The food was not good.	−0.3412
The class was not boring.	**+0.2411**

If “not,” “never,” or a word like “isn’t” comes within **3 words** before, the score is multiplied by **−0.74**.

Check: 1.9 × −0.74 = −1.406, and −1.406 ÷ √(1.406² + 15) = **−0.3412**.

RULE 5

“But” shifts the focus

Sentence	Compound score
The food was good, but the service was horrible.	`−0.5859`
The service was horrible, but the food was good.	**`+0.3818`**

Words **before** “but” count half (× 0.5). Words **after** “but” count more (× 1.5).

Same words, different order, opposite result! The part after “but” is usually what the writer really means.

SOCIAL MEDIA

Emojis, emoticons, and slang

Sentence	Compound score
The food was good.	`0.4404`
The food was good :)	**`0.7096`**
The food was good 😍	**`0.7096`**
The food was sooo good lol	**`0.6908`**

VADER's lexicon includes emoticons like :) and slang like "lol." It also turns emojis into words, so 😍 becomes "smiling face with heart-eyes."

SIMULATED DEMO

Watching the rules work

SIMULATED DEMO

```
sentences = ["The food was good.",
             "The food was GOOD!!!",
             "The food was not good.",
             "Wow, this is the BEST day ever!!! 😊"]

for s in sentences:
    c = analyzer.polarity_scores(s)["compound"]
    print(f"{c:+.4f}  {s}")
```

OUTPUT

```
+0.4404  The food was good.
+0.6714  The food was GOOD!!!
-0.3412  The food was not good.
+0.9486  Wow, this is the BEST day ever!!! 😊
```

Each rule changed the score in the direction we predicted.

The last sentence uses almost every rule at once and gets a very high score: **+0.9486**.

In the notebook, you will test each rule one at a time.

04

Build Your Own Mini-VADER

Implementing the rules in Python.

BUILD IT

The plan for our mini-VADER

1 — Lexicon
A small dictionary of word scores

→

2 — Find words
Look up each word

→

3 — Apply rules
CAPS, very, not, but, !

→

4 — Add up
Total of all scores

→

5 — Squeeze
total ÷ √(total² + 15)

Our mini-VADER uses only **9 words** and the 5 rules, in about 30 lines of Python.

BUILD IT

The heart of the code

SIMULATED DEMO

```
LEXICON = {"good": 1.9, "great": 3.1, "love": 3.2,
           "bad": -2.5, "hate": -2.7, ...}
BOOSTERS = {"very": 0.293, "slightly": -0.293, ...}
NEGATIONS = {"not", "never", "isn't", ...}

for i, word in enumerate(clean):
    score = LEXICON[word]
    if previous word in BOOSTERS:  score += 0.293
    if word is CAPS:               score += 0.733
    if a negation is near:         score *= -0.74

total += 0.292 * number_of_exclamations
compound = total / math.sqrt(total**2 + 15)
```

This is a **simplified** view of the code. The notebook has the full, working version with comments.

Each rule is just one or two lines of Python.

The real VADER has more rules, but these five do most of the work.

TEST IT

Our mini-VADER vs. the real VADER

We tested our mini-VADER on 14 sentences. It matched the real VADER every time. Here are six of them.

Sentence	Our mini-VADER	Real VADER
The food was good.	+0.4404	+0.4404
The food was GOOD!!!	+0.6714	+0.6714
The food was not very good.	−0.3865	−0.3865
The service was horrible, but the food was good.	+0.3818	+0.3818
The show was really amazing!!	+0.6885	+0.6885
The plot was okay, but the ending was amazing.	+0.7684	+0.7684

For words outside our 9-word dictionary, our version would fail. That is why the real lexicon has 7,506 entries.

05

VADER on Real Data

When does it work, and when does it struggle?

REAL DATA

Two real datasets

10,000 tweets

5,000 positive and 5,000 negative tweets from the NLTK library. About 99% of them contain smileys like :) or :(.

10,662 movie review snippets

Short sentences from movie critics, half positive and half negative. They use formal language and few emojis.

We check VADER's label against the true label. If VADER says "neutral," we count it as wrong.

RESULTS

VADER vs. a trained model

How often was VADER correct? We also compared it with a TF-IDF model trained on 75% of the same data (Lesson 2 style), tested on the other 25%.

Tweets



Movie review snippets



Scores are accuracy on the same 25% test set: tweets 79.2% vs. 76.4%, movie snippets 53.6% vs. 76.9%.

RESULTS

Why such different results?

Why VADER wins on tweets

Tweets are short and full of emojis, smileys, slang, and “!!!”. That is exactly what VADER was built for. Our TF-IDF settings also threw away the smileys.

Why VADER loses on movie reviews

Critics write in a subtle, formal way. 1,823 of the snippets had no strong feeling words, so VADER called them neutral.

Example: “to portray modern women the way director davis has done is just unthinkable.” The true label is **negative**, but VADER scored it **0.000**.

06

Limits of VADER

Rules cannot catch everything.

LIMITS

Where VADER makes mistakes

Sentence	VADER's score
Oh great, another Monday.	+0.6249
The food was gud	0.0000
Sobrang sarap ng pagkain!	0.0000

Sarcasm: "great" sounds positive, but the writer is actually annoyed.

Misspellings: "gud" is not in the lexicon, so it scores zero.

Other languages: VADER only knows English. "Sarap" means delicious!

SIMULATED DEMO

Improving VADER: teach it new words

SIMULATED DEMO

```
# Teach VADER some Filipino words
analyzer.lexicon.update({"sarap": 2.5, "masarap": 2.5,
                         "ganda": 2.5, "pangit": -2.5})

for s in ["Sobrang sarap ng pagkain!",
          "Ang ganda ng movie!",
          "Ang pangit ng ending."]:
    print(analyzer.polarity_scores(s)["compound"], s)
```

OUTPUT

```
0.5848 Sobrang sarap ng pagkain!
0.5848 Ang ganda ng movie!
-0.5423 Ang pangit ng ending.
```

Because VADER is just a dictionary and rules, we can **add our own words**.

Before the update, all three sentences scored 0. Now VADER understands them.

But the rules are still English. “Pero” does not trigger the “but” rule, and “sobrang” is not a booster yet. In the notebook, you will fix that too.

BIG PICTURE

VADER vs. the other methods you know

Feature	VADER	Trained model (Lesson 2)	Transformer (Lesson 4)
Training data needed?	**No**	Yes	Yes (pre-trained)
Speed	**Very fast**	Fast	Slower
Easy to explain?	**Yes, every point is traceable**	Somewhat	Hard
Understands context?	Only simple rules	No	**Yes**
Best for	Social media, quick checks	One specific dataset	Hard, subtle text

There is no single best tool. A good engineer picks the one that fits the data and the job.

SUMMARY

Key takeaways

1. Sentiment analysis finds out if a text is positive, negative, or neutral.
2. VADER uses a dictionary of 7,506 rated words plus rules.
3. The compound score is total ÷ √(total² + 15), from -1 to +1.
4. Exclamation points, CAPS, and boosters make feelings stronger.
5. “Not” flips the feeling, and the part after “but” matters more.
6. VADER is great for short social media text but weaker on formal reviews.
7. VADER can be improved by adding new words, like Filipino words.

CHECK YOUR UNDERSTANDING

Quick quiz

Q1

What does VADER need before it can work?

A. Thousands of labeled examples
B. Nothing; its dictionary and rules are ready
C. A GPU

Q2

A compound score of -0.62 means the text is…

A. Positive
B. Neutral
C. Negative

Q3

What happens to “good” in “not good”?

A. Its score is multiplied by -0.74
B. It is removed
C. It becomes stronger

Q4

Which part counts more in “I liked it, but it was too long”?

A. “I liked it”
B. “it was too long”
C. Both count the same

Q5

Why did VADER score “Oh great, another Monday.” as positive?

A. It cannot detect sarcasm
B. Mondays are positive
C. It made a math error

CHECK YOUR UNDERSTANDING

Quiz answers

Q1 **Answer: B**
VADER works right away with no training.

Q2 **Answer: C**
Anything at or below -0.05 is negative.

Q3 **Answer: A**
Negation flips and weakens the feeling.

Q4 **Answer: B**
Words after “but” count 1.5 times as much.

Q5 **Answer: A**
It only sees the word “great,” not the annoyed tone.

ACTIVITY

Try it yourself

Activity 1: Human VADER

Give groups five sentences. They calculate the compound score by hand using the lexicon and rules, then check with Python.

Activity 2: Taglish lexicon

Each group adds 10 Filipino or Taglish words with scores from -4 to +4. Test them on real class comments.

Activity 3: Fool VADER

Write sentences that VADER gets wrong, like sarcasm. Explain which rule or missing word caused each mistake.

Bonus: Use VADER to analyze anonymous feedback about a school event and present the results in a chart.

NATURAL LANGUAGE PROCESSING

Thank you for listening!

You can now measure feelings in text and build your own sentiment tool.

What questions do you have?

`this` `lesson` `was` `GREAT` `!!!` `:)`